

Research paper

Multi-objective generative design framework and realization for quasi-serial manipulator: Considering kinematic and dynamic performance

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ABSTRACT

This paper proposes a framework for optimizing the linkage mechanism of a quasi-serial manipulator for target tasks. The process is illustrated through a case study of a two-degree-of-freedom (2-DOF) linkage mechanism, which significantly influences the workspace of the quasi-serial manipulator. First, a diverse set of quasi-serial mechanisms is generated with workspaces that satisfy the target task and is converted into three-dimensional computer-aided design (3D CAD) models. Then, kinematic and dynamic analyses are conducted to compute workspace and payload torque labels for surrogate model training. Adaptive sampling is employed to identify an appropriate amount of training data for accurate prediction, while ensemble models are utilized to enhance robustness. After model training, a multi-objective optimization problem is formulated under the mechanical and dynamic constraints of the manipulator. The design objective is to recommend quasi-serial mechanisms with optimized kinematic (workspace) and dynamic (payload torque) performance that fulfill the target task. To explore the underlying physics of the Pareto solutions obtained via the Non-Dominated Sorting Genetic Algorithm (NSGA-II), various data mining techniques are applied to extract design rules that offer practical guidance. Finally, a detailed manipulator is realized using 3D-printed parts with topology optimization, and its performance is verified through a payload test. These results demonstrate the potential of the proposed framework for broader mechanism applications and its capability to support practical design decisions through design rule extraction.

1. Introduction

Designing kinematic mechanisms is an important topic covered in a wide range of areas across the industry. Depending on the application, the mechanism is classified into a passive and actuating system (Li et al., 1996; Shao et al., 2016; Stroppa, 2024; Portilla-Flores et al., 2011). For example, the passive system (Raghavan, 2004; Wongrataphisan and Cole, 2008) can be classified as automobile suspension, compensation mechanism, and the actuating system (Bergelin and Voglewede, 2012; Deshpande and Purwar, 2017; Kim et al., 2008) can be classified as a robot gripper, manipulator mechanism. Passive system design is mainly a structure required for fixed external conditions. Therefore, it is relatively straightforward to intuitively define design elements for optimization. In contrast, the actuation system also needs to consider the changing external conditions due to the degree of freedom. Therefore, it is significantly challenging to intuitively define design problems for optimization. Typically used as a power source in an actuating system, the motor can only perform circular motion.

However, tasks in real-world applications or industries frequently require variable workspaces. The linkage mechanism is commonly used because it is inefficient to include the task region with only circular motion. The linkage mechanism has the potential to perform various tasks because it has an attractive feature that can transmit circular motion to different path motions (Hassan and Abomoharam, 2017; Heyrani Nobari et al., 2022; Kim et al., 2016; Li et al., 2022; Sedlaczek and Eberhard, 2009; Shao et al., 2016; Wang and Lan, 2014; Yim et al., 2019).

The linkage mechanism transmits the torque from the drive shaft to the end-effector. The rotation of the drive shaft is transmitted to an end-effector workspace by kinematic motion. Workspace means the space that the end-effector can reach. Torque transmission means the continual torque transfer from the drive shaft to the end-effector and is highly related to the payload. The linkage mechanism is characterized by having a significantly sensitive workspace according to the change in link length. In addition, since the mechanical changes caused by

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this significantly impact torque transmission, designing an appropriate mechanism remains a challenging problem. Existing design strategies to detour these problems use the method of conceiving a mechanism that satisfies the workspace and selecting an actuator that provides the required torque performance. However, these methods have significant problems in terms of optimization. First, in many cases, linkage mechanisms of the workspace that can perform tasks are not unique if there is enough design space. Therefore, the linkage mechanism optimized with the required actuator torque was not considered. Secondly, if there is no actual linkage to match the target workspace, the optimization iteration can fail to converge. Finally, the workspace and torque transmission are highly sensitive to changes in linkage length (Lee et al., 1999; Tsai and Lai, 2004). Therefore, simultaneously optimizing workspace and torque transmission requires significant feedback cycles or wide-range thresholds (Barnawal et al., 2017; Safoutin and Smith, 1998; Schütze et al., 2003).

For these reasons, various numerical analysis frameworks have been proposed in classical mechanism design to enable iterative optimization. For instance, one approach divided the design space into discrete mesh blocks and synthesized linkages by minimizing the deformation energy of virtual zero-length springs within each block (Yu et al., 2020; Kim et al., 2007; Yim et al., 2019). This method allows high-precision synthesis of linkages that follow kinematically feasible target paths. In this study, one notable case demonstrated near-perfect tracking of the target path at the 220th iteration. However, it can suffer from convergence issues, especially when applied to artificially generated workspaces. Furthermore, since the target path is used as the sole objective function, this method inherently neglects torque transmission, which is critical for actuation. In cases where the desired path cannot be synthesized within the potential of a four-bar linkage, the iterative process may diverge without convergence. Another study proposed optimizing the structure of a gripper composed of a linkage mechanism to maximize holding force (Hassan and Abomoharam, 2017). Dynamic models were numerically formulated and optimized in that work to enhance grip performance. However, the workspace of the gripper was not considered, which limits the generality of the design. These numerical-based methodologies typically yield unique solutions through a forward modeling approach. However, in many real-world scenarios, multiple linkages can satisfy the same task conditions, suggesting the need for alternative strategies that allow the exploration of diverse feasible designs. To improve computational efficiency, significant iterative optimization studies assume a rigid body by ignoring mass and inertia. This simplification enables rapid numerical estimation of geometric responses and facilitates repeated solution searches. For example, several studies numerically calculated dynamic responses under specific geometric constraints or iteratively evaluated task-based configurations (Barnawal et al., 2017; Safoutin and Smith, 1998; Schütze et al., 2003). However, these numerical approaches require explicit deterministic formulations of all governing relationships, which limits their applicability as the system complexity increases. In high-dimensional engineering problems, simulation-based analyses are often essential to obtain accurate system responses. When such analyses are required at every iteration, the computational cost grows significantly, making conventional methods less effective for complex mechanism design.

To address these limitations more effectively, this study adopts a surrogate-based optimization approach. While surrogate-based engineering design has been actively investigated across various domains, most studies have focused on parametric optimization of structural performance for individual components (Yang and Yee, 2022; Yang et al., 2023; Yim et al., 2021; Heyrani Nobari et al., 2022). In mechanism synthesis, one notable study proposed a data-driven method that utilized 100,000 samples to identify linkages that satisfy a given motion (Yim et al., 2021). Another study introduced a massive dataset of 1.1 billion samples to improve model training for target path tracking, claiming 800 times faster than traditional simulation-based methods (Heyrani Nobari et al., 2022). These approaches often rely on

the simplified assumptions of early-stage mechanism design, such as idealized rigid bodies and the neglect of mass and inertia. However, in real-world engineering problems, collecting such large datasets is often infeasible, especially when system responses require high-cost simulations. Thus, when addressing practical engineering problems that involve system responses requiring numerical analysis through engineering software, it is essential to consider the computational cost of generating labeled data. Rather than relying on too large datasets, it becomes critical to construct robust surrogate models using a sufficiently feasible number of design samples. This study highlights the limitations of prior approaches, which focused solely on rigid linkage mechanisms under kinematic assumptions while neglecting mass and inertia. In contrast, the proposed framework incorporates dynamic performance using three-dimensional (3D) simulations and introduces computational strategies to minimize simulation cost without compromising model fidelity.

To apply a practical optimization scenario that requires simultaneous consideration of kinematic and dynamic performance, an industrial quasi-serial manipulator is selected as the application case. This manipulator configuration has been validated for high-payload environments and is known to provide enhanced load-bearing capacity and structural robustness compared to traditional serial architectures (Klimchik et al., 2016b; Klimchik and Pashkevich, 2017; Klimchik et al., 2016a). To design a quasi-serial manipulator suitable for high-payload environments, this study adopts a four-bar linkage mechanism, specifically applying the crank-rocker configuration. While industrial manipulators typically require at least six degrees of freedom (DOF) to perform general tasks in 3D space, this study simplifies the problem by focusing on the dominant 2-DOF that primarily determines the reachable workspace. This simplification aims to validate the scalability and applicability of the proposed surrogate-based optimization framework for task-specific mechanism design. Based on the resulting optimal concept design variables, an additional 4-DOF, less sensitive to workspace constraints, is included in the detailed design stage. In this process, topology optimization is applied to realize the final structure of the mechanism, thereby demonstrating the feasibility of the proposed framework in real-world applications.

The design of engineering applications involving kinematic mechanisms frequently requires kinematic analysis and dynamic or even more complex mechanical evaluations. Solving such practical engineering problems typically relies on simulation, and optimizing multiple system responses (such as kinematic and dynamic performance) can incur substantial computational costs. Therefore, rather than relying solely on high-fidelity simulations or excessively large datasets, this study emphasizes the importance of constructing a feasible yet robust surrogate model through a well-balanced approach. To this end, adaptive sampling is employed to selectively enrich the dataset with informative samples, and ensemble learning is used to enhance model stability and accuracy. This approach significantly reduces computational effort while maintaining high predictive performance, enabling an efficient and scalable optimization framework. The proposed generative framework facilitates the discovery of optimal mechanism designs and can be extended to other mechanism-based design problems. The main contributions of this framework are summarized as follows:

1. To the authors' best knowledge, this is the first attempt to consider kinematic and dynamic performance simultaneously for quasi-serial manipulator mechanism design problems.
2. To overcome the limitations of existing methods of optimizing kinematic performance, generative modeling is adopted to generate numerous task-based mechanisms considering kinematic workspace. Which has been used as a method to actively design mechanisms and can also be used in other applications.
3. To solve real engineering problems, a multi-objective function is usually required. In this study, 3 objective functions were selected to optimize the quasi-serial manipulator, and constraints were also considered to find the Pareto solution.

4. To address real-world engineering problems, this study formulates a multi-objective optimization (MOO) problem by defining three objectives and solving it using adaptive sampling, ensemble modeling, and the non-dominated sorting genetic algorithm II (NSGA-II) algorithm. The final design solution, when implemented, demonstrated over 85% reduction in total design time compared to traditional iterative analysis-based design approaches.
5. Design rules obtained through data mining offer practical guidance, enabling early feasibility evaluation and efficient iteration in mechanism system design.

Section 3 describes the proposed framework and its advantages. Section 4 presents the results, including a quantitative ablation study comparing non-adaptive, adaptive sampling, NSGA-II, and sequential quadratic programming (SQP) optimization methods, and Section 5 provides a sensitivity analysis for the data mining process. Section 6 details the manufacturing and experimental validation of the recommended optimal quasi-serial manipulator. Finally, Section 7 summarizes the findings and key contributions of the study.

2. Related work

2.1. Surrogate modeling in engineering design

Surrogate modeling has been widely adopted in engineering design to reduce computational costs associated with expensive simulations (Yang and Yee, 2022; Regenwetter et al., 2023; Lee and Kang, 2025). Several studies have employed big data approaches in mechanism synthesis to generate and recommend linkage candidates that satisfy kinematic constraints (Yüksel et al., 2023; Heyrani Nobari et al., 2022; Deshpande and Purwar, 2019). Other approaches have utilized the synthesized linkage workspace and mechanism images to retrieve solutions similar to the target workspace through data-driven methods (Deshpande and Purwar, 2021). However, acquiring a sufficiently large and diverse dataset to train accurate surrogate models remains a persistent challenge. Moreover, these studies primarily focused on geometric or kinematic conditions, without incorporating dynamic performance or other critical engineering analyses in detailed design stages. Thus, extending surrogate modeling to mechanism design, which involves optimizing dynamic responses and multiple interacting components, remains challenging due to the complexity of geometric and dynamic coupling.

2.2. Adaptive sampling for data efficiency

Gaining sufficient data for accurate surrogate modeling frequently incurs prohibitive computational costs, especially in high-dimensional design problems. To address the risks of overfitting caused by excessive data and prediction failures due to insufficient data, incremental model updating strategies have been explored (Gorissen et al., 2010; Jiao et al., 2024; Hong et al., 2024). Adaptive sampling techniques selectively add new data points to better explore the design space or target regions with high prediction uncertainty, thereby improving model accuracy and minimizing the number of required simulations. Recent studies have demonstrated the effectiveness of adaptive sampling even in physics-informed neural networks (PINNs). For example, a framework employing Gaussian adaptive sampling (GAS) has been proposed to enhance the accuracy of PINNs when modeling functions that can be defined by governing equations (Jiao et al., 2024). In addition, adaptive sampling has been applied in manufacturing control systems facing expensive MOO problems by mapping high-dimensional objective spaces onto two-dimensional radial spaces to improve surrogate model generalization (Hong et al., 2024).

2.3. Ensemble learning for robust surrogate modeling

Even when a surrogate model demonstrates high predictive accuracy, it can produce inconsistent predictions under new data or slightly different hyperparameters, especially when trained on sensitive and localized regions of the design space. To address this issue and enhance model robustness, ensemble learning techniques have been introduced (Dong et al., 2020; Zheng et al., 2025; Dietterich, 2000). An ensemble model is constructed by aggregating the predictions of multiple independently trained models, thereby reducing variance and improving generalization performance. However, ensuring that each base model contributes meaningfully to the ensemble requires careful tuning of its configuration. To automate and optimize this process, automated machine learning (AutoML) techniques are often employed to optimize the hyperparameters of individual models before aggregation (He et al., 2021; Snoek et al., 2012). In engineering optimization, ensemble-based surrogates have been shown to provide more stable and reliable predictions across diverse design spaces compared to single-model approaches (Wen et al., 2025; Regenwetter et al., 2023). This is particularly critical in mechanism design applications, where dynamic responses can be susceptible to minor geometric variations.

2.4. Multi-objective optimization

Optimization techniques for engineering performance often employ heuristic methods such as genetic algorithms (GAs) (Mirjalili and Mirjalili, 2019; Lambora et al., 2019) or numerical approaches like SQP (Boggs and Tolle, 1995), an extension of the Newton–Raphson method. While these methods are well-established for single-objective optimization, many engineering design problems inherently involve multiple conflicting objectives, necessitating MOO techniques to identify trade-off solutions. Among various MOO algorithms, the NSGA-II has become one of the most widely used due to its computational efficiency and ability to maintain a diverse set of Pareto-optimal solutions (Deb et al., 2002). However, because GA-based optimization relies on heuristic search processes, the obtained solutions may differ from those derived using purely numerical optimization methods. Therefore, when the surrogate model's outputs are sufficiently continuous regarding the design variables, it is advisable to validate the solution ranges using numerical optimization methods to ensure the reliability and accuracy of the selected optimal designs.

3. Framework

The ML-aid optimal linkage mechanism framework proposed in this study consists of four stages, and its overall flowchart is shown in Fig. 1. First, it involves building an automation process aimed at generating and analyzing mechanisms capable of performing tasks through generative modeling and analysis (Stages 1 to 2). A surrogate model is then trained to predict the performance of the generated 3D computer-aided design (CAD) model, incorporating adaptive sampling to enhance the training process and utilizing an ensemble modeling approach to improve prediction robustness and accuracy (Stage 3). Finally, NSGA-II is applied to the already trained surrogate model, recommending optimal concept mechanisms (Stage 4) (Deb et al., 2002). The following sections are organized to demonstrate each stage of the proposed framework in detail.

3.1. Stage 1: Kinematic mechanism generation

Stage 1 includes creating task-based quasi-serial mechanisms and converting them into 3D CAD models. As shown in Fig. 2, the task consists of the target region and payload. To approach a problem in general, it is important to define a task region that can contain all task points. While there are various methods to express the boundaries of

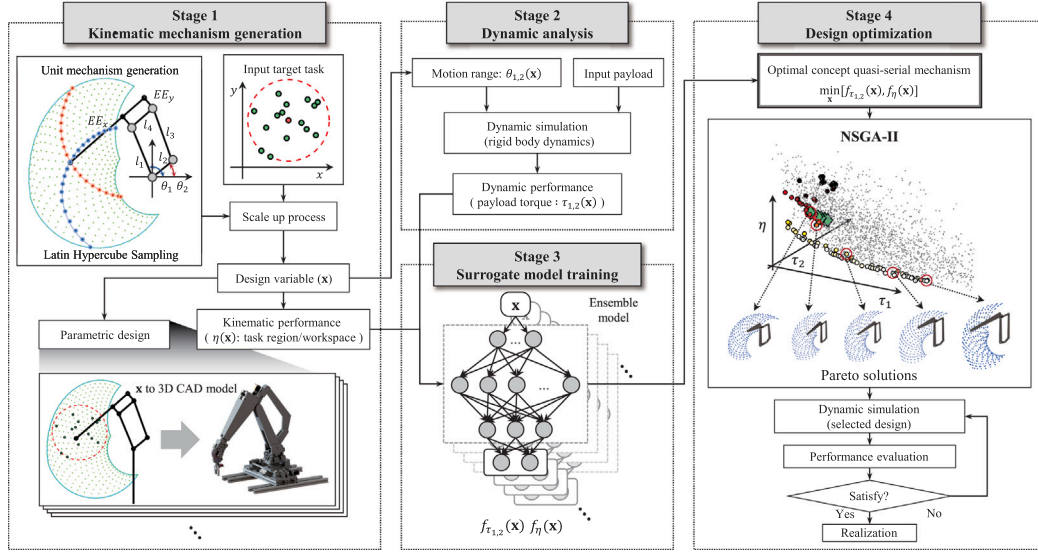


Fig. 1. Multi-objective generative mechanism design framework.

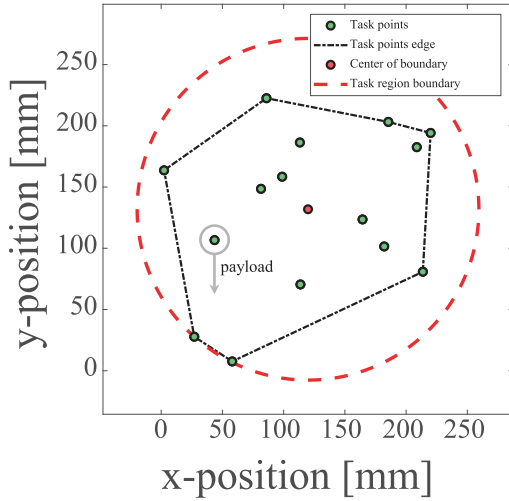


Fig. 2. Example of target task.

the task points, this study defines a minimum circle containing all task points as a workspace to make the problem relatively easily accessible.

The quasi-serial mechanism is a 2-degrees of freedom (DOF) linkage consisting of a frame (l_1), crank (l_2), coupler (l_3), rocker (l_4), upper arm (EE_x, EE_y), and end-effector, as shown in Fig. 3(a). Workspace is a reachable space of the end-effector according to the motion of the frame (θ_1) and crank (θ_2).

The quasi-serial mechanism generation with workspaces including target regions is summarized in the following process (in Fig. 3). The length of the unit mechanism is determined by assuming the length of the frame (l_1) as 1. Then, the length of each link, including l_2, l_3, l_4, EE_x , and EE_y , is defined as a relative ratio value to the frame length l_1 . The various designs based on the defined link parameters are generated using Latin hypercube sampling (LHS), and the range of each design parameter is shown in Table 1.

When the generated designs do not satisfy the condition of crank-rocker mechanisms (Eq. (1)), they are filtered out.

$$l_3 + l_2 < l_1 + l_4 \quad (1)$$

From the process described above, 200,000 samples were generated, and after filtering, 11,080 LHS samples were obtained that satisfy

Table 1
Length range of the link parameter.

x	l_1	l_2	l_3	l_4	EE_x	EE_y
min	1	0.18	0.8	0.3	1.0	0.2
max	1	0.6	1.3	0.6	1.4	0.7

the requirements for unit quasi-serial mechanisms. At this stage, the operating range is defined as $45^\circ \leq \theta_1 \leq 180^\circ$ and $-37.5^\circ \leq \theta_2 \leq \theta_1$. The workspace of each generated unit mechanism undergoes a scale-up process to include the task region. This scale-up process is generally performed to address task-dependent design requirements. Task regions are incorporated by multiplying unit mechanisms by an appropriate scale factor, which is defined as the minimum scale required for the workspace of the unit mechanism to encompass the task region. This factor can be fine-tuned based on a threshold to account for design safety considerations. Consequently, the quasi-serial mechanism is determined as the product of the unit mechanism and the scale factor.

The kinematic performance of the generated quasi-serial mechanism is defined as η = target region/workspace. The closer the value of η is to 1, the more it means that only the target task can be included. However, since this performance only considers kinematics but not dynamics. Therefore, to take into account the dynamic performance, the generated quasi-serial mechanism is converted to a 3D CAD model by Rhinoceros 3D® software to perform dynamic analysis as in Fig. 3(b) (McNeel et al., 2020).

3.2. Stage 2: Dynamic analysis

Stage 2 includes a dynamic simulation process to obtain the dynamic performance metrics used as labels for training the surrogate model in Stage 3. The vast set of quasi-serial mechanisms generated in Stage 1 satisfied the target workspace conditions based solely on kinematic performance. Since Stage 1 focused only on kinematic aspects, Stage 2 aimed to evaluate the dynamic performance of these mechanisms.

Without considering actuator performance, all mechanisms generated in Stage 1 could be considered feasible solutions (Kim et al., 2016; Shao et al., 2016; Hassan and Abomoharam, 2017). However, from an optimization and efficiency perspective, this approach is inappropriate. Design optimization using surrogate models is essential to identify mechanisms that balance kinematic and dynamic performance.

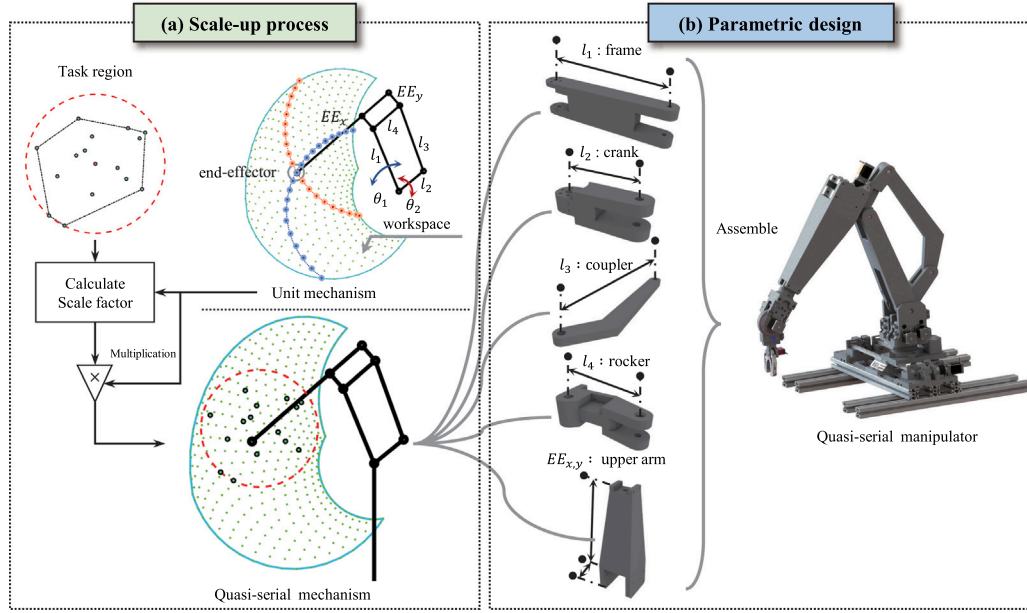


Fig. 3. Quasi-serial manipulator generation process.

While the geometric behavior of simple mechanisms can be evaluated through direct mathematical calculations (Hassan and Abomoharam, 2017; Wang and Lan, 2014; Bergelin and Voglewede, 2012), applying such physics-based modeling becomes significantly challenging when multiple mechanical components are assembled, and dynamic interactions must be considered. Therefore, in this study, performance labeling was conducted through simulation-based analysis.

Dynamic simulations using RecurDyn 2023 (FunctionBay FunctionBay, 2023) with rigid body dynamics (RBD), A payload of 5 kg was applied along the EE_x direction at the end-effector, representing a typical operational scenario where a gripper holds an object during task execution. The frame (l_1) and crank (l_2) were rotated based on geometric relations defined by the design variables in Stage 1, and the corresponding rotation angles $\theta_{1,2}$ were automatically calculated. The required joint torques ($\tau_{1,2}$) were obtained through the simulation and utilized as dynamic performance labels for surrogate model training. The rigid body assumption was adopted instead of full multi-flexible body dynamics (MFB) analysis to ensure rapid evaluation while maintaining sufficient physical fidelity for the targeted optimization framework.

3.3. Stage 3: Surrogate model training

This stage focuses on training surrogate models to efficiently predict the kinematic and dynamic performances of quasi-serial mechanisms. Given the geometric complexities inherent in these designs, which introduce significant non-linearities in both kinematic and dynamic behaviors, selecting an appropriate surrogate modeling approach is crucial.

While various surrogate modeling methods, such as Gaussian process regression (GPR), Support vector machines (SVM), and Support vector regression (SVR), can be employed in engineering design, this study adopts a multilayer perceptron (MLP) model. Although methods like GPR can provide prediction uncertainties, in the proposed application (where RBD analysis is primarily used), it is more reliable to select Pareto solutions by simulating those obtained from the surrogate model rather than relying on uncertainty estimates. MLP was chosen as it is compatible with advanced deep learning techniques and allows gradient-based optimization, offering greater flexibility and scalability for future extensions of the framework.

In recent studies related to mechanism synthesis, various deep learning approaches such as convolutional neural networks (CNNs), conditional generative adversarial networks (cGANs), and conditional variational autoencoders (cVAEs) have been explored, mainly focusing on generating linkage configurations that match target paths or geometries (Lee et al., 2024; Deshpande and Purwar, 2019; Yim et al., 2021; Heyrani Nobari et al., 2022). However, their lack of consideration for dynamic responses becomes a critical limitation in applications requiring additional engineering evaluation. To address these challenges and ensure that the model captures meaningful physical behavior while maintaining robustness, this study adopts a multilayer perceptron (MLP) as the base surrogate model. The MLP architecture is highly effective for low-dimensional regression problems and offers better interpretability than complex generative models. Given the nature of the proposed application — predicting scalar-valued performance metrics (e.g., η , τ_1 , τ_2) from 1D design variables — the MLP was determined to be an appropriate and reliable choice for surrogate modeling (Goodfellow et al., 2016; Jahirul et al., 2021; Meng and Karniadakis, 2020; Yang and Yee, 2023; Yang et al., 2023).

In this stage, the surrogate model is trained following the workflow illustrated in Fig. 5. The process begins with constructing the train dataset using the design variables and performance labels obtained from the dynamic simulations in Stage 2. Initially, a sparse dataset is generated to minimize computational costs. Using this dataset, AutoML is employed to identify the optimal hyperparameters for the MLP model (He et al., 2021; Snoek et al., 2012). This step does not involve adaptive sampling but serves as a baseline to establish a robust and efficient initial configuration of the model. These hyperparameters act as reference settings for subsequent model updates during adaptive sampling, ensuring faster convergence and improved performance.

The MLP is implemented with an input layer, multiple hidden layers, and an output layer. For a given input $\mathbf{x} \in \mathbb{R}^d$, the network is structured as follows:

$$h^{(1)} = \text{ReLU}(W^{(1)}\mathbf{x} + b^{(1)}),$$

$$h^{(l)} = \text{ReLU}(W^{(l)}h^{(l-1)} + b^{(l)}), \quad l = 2, 3, \dots, L,$$

$$\mathbf{y} = W^{(L+1)}h^{(L)} + b^{(L+1)}, \quad (2)$$

where $W^{(l)}$ and $b^{(l)}$ represent the weights and biases for each layer l , and L is the total number of hidden layers. The loss function for

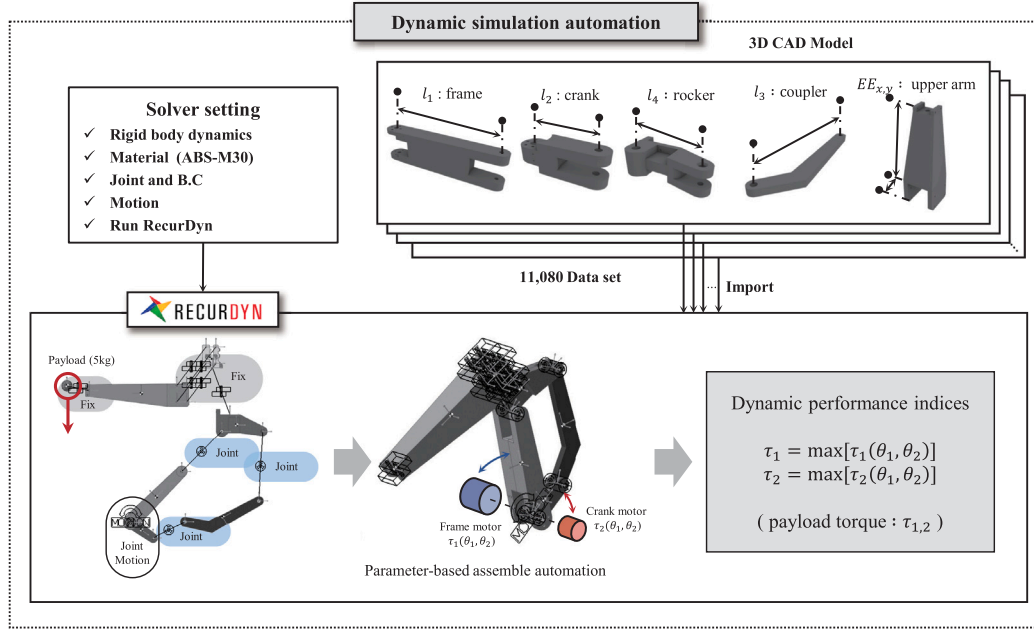


Fig. 4. Dynamic simulation automation process.

training the MLP is defined as the mean squared error (MSE):

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \left(\mathbf{y}_{\text{true}}^{(i)} - \mathbf{y}^{(i)} \right)^2. \quad (4)$$

The hyperparameters of the MLP, including the number of hidden layers L , the number of neurons per layer, and the learning rate, are determined using the AutoML process. AutoML evaluates a range of model configurations and optimizes these parameters to balance model complexity and generalization ability, using the initial training dataset constructed from design variables and performance labels obtained in Stage 2. This step establishes a baseline model configuration, which serves as the foundation for subsequent adaptive sampling and model refinement.

The surrogate model, denoted as $\mathcal{M}$, is implemented using a MLP to approximate the relationship between design variables ($\mathbf{x}$) and performance outputs ($\mathbf{y}$). $\mathcal{M}$ represents a parameterized mapping function, where the model parameters are optimized during the training process. This mapping is defined as:

$$\mathcal{M} : \mathbf{x} \mapsto \mathbf{y}, \quad (5)$$

where $\mathbf{x}$ is the input vector of design variables, and $\mathbf{y}$ is the predicted performance metrics. The model's uncertainty, $\mathcal{U}(\mathbf{x}; \mathcal{M})$, quantifies the confidence in predictions made by $\mathcal{M}$ and is typically derived from the variance of predictions across different regions of the design space.

Training a highly accurate surrogate model requires careful consideration of the design space and sampling strategy. While conducting a dense design of experiments (DoE) across the entire design space might theoretically yield a superior model, such an approach is often impractical due to the high computational cost of high-fidelity simulations and the potential risk of overfitting to redundant data. To address these challenges, an adaptive sampling approach is employed to iteratively refine the training dataset by selectively adding samples in regions where the model's uncertainty is highest:

$$\mathbf{x}_{\text{new}} = \arg \max_{\mathbf{x}} \mathcal{U}(\mathbf{x}; \mathcal{M}), \quad (6)$$

where $\mathcal{U}(\mathbf{x}; \mathcal{M})$ represents the model's uncertainty, typically quantified by the prediction variance. The updated dataset is defined as:

$$D_{\text{train}} \leftarrow D_{\text{train}} \cup \{(\mathbf{x}_{\text{new}}, \mathbf{y}_{\text{new}})\}. \quad (7)$$

The AutoML-derived hyperparameters guide the adaptive sampling process by ensuring that the surrogate model remains well-tuned during iterative updates. If the surrogate model fails to achieve satisfactory performance after adding new samples, the AutoML process is revisited to re-optimize the hyperparameters based on the updated dataset. This iterative combination of AutoML and adaptive sampling enhances model robustness and predictive accuracy while avoiding excessive computational overhead.

After the adaptive sampling process stabilizes and the training dataset becomes sufficiently comprehensive, the surrogate model undergoes an ensemble modeling phase. In this phase, multiple surrogate models are trained independently using the refined dataset, with each model configured slightly differently to capture diverse patterns in the data. The predictions from these individual models are aggregated into a single output using a weighted average based on their R^2 scores (Dong et al., 2020; Dietterich, 2000):

$$\hat{\mathbf{y}}_{\text{ensemble}} = \sum_{k=1}^K w_k \hat{\mathbf{y}}^{(k)}, \quad w_k = \frac{R_k^2}{\sum_{j=1}^K R_j^2}, \quad (8)$$

where K is the total number of models, R_k^2 is the coefficient of determination for the k th model, and w_k is the weight assigned to each model.

The proposed training process includes an ensemble modeling approach, which has been widely utilized for its ability to enhance the robustness and reliability of predictive models. By aggregating outputs from multiple models, the ensemble approach mitigates biases and uncertainties associated with individual models. In this stage, ensemble modeling is employed to improve the stability and accuracy of surrogate models specifically developed for mechanism design tasks. This ensures that the trained models remain resilient to variations in design parameters and data quality. To achieve this, the process integrates three key components: AutoML for optimizing hyperparameters, adaptive sampling for iteratively updating the dataset, and ensemble modeling for enhancing overall robustness. AutoML determines the optimal model configuration by balancing complexity and generalization, while adaptive sampling expands the training dataset by selectively sampling high-uncertainty regions. Ensemble modeling minimizes discrepancies among individual models by applying performance-based weights, thereby consolidating predictions into a single, robust output.

3.4. Stage 4: Design optimization

This stage focuses on optimizing kinematic and dynamic performances, which are the outputs of the trained ensemble model from Stage 3. Since the optimization process requires numerous evaluations, the trained surrogate model is utilized to efficiently evaluate performance. A MOO problem is formulated, as this study considers three objective functions: f_η , f_{τ_1} , and f_{τ_2} . The well-known NSGA-II algorithm is employed as the optimizer, and for this purpose, the Python package `PyMOO` is used (Blank and Deb, 2020). The objective function, constraints, and settings for NSGA-II are defined as follows:

3.4.1. Design variables

The six design variables are $\mathbf{x} = [l_1, l_2, l_3, l_4, EE_x, EE_y]$, where all of the elements are the quasi-serial mechanism design variables used in Section 3.1 (Fig. 3).

3.4.2. Objective functions and constraints

$f_\eta(\mathbf{x})$ represents the predicted kinematic performance, where $\eta(\mathbf{x})$ was defined in Stage 1 as a function of $\mathbf{x}$. A lower value of η indicates a larger workspace for performing the target task, which also enhances flexibility for additional tasks. In Stage 3, the predicted kinematic performance from the ensemble model is similarly defined as a function of $\mathbf{x}$ and is utilized as an objective function to minimize. The terms $f_{\tau_1}(\mathbf{x})$ and $f_{\tau_2}(\mathbf{x})$ represent the predicted joint torques required to actuate the quasi-serial manipulator. These values are also used as objective functions to minimize the actuator torques for performing the same target task. Ultimately, $(f_\eta, f_{\tau_1}, f_{\tau_2})$ are selected as objective functions, with all three set to be minimized.

$$\begin{aligned}
 &\text{find} && \mathbf{x} = [l_1, l_2, l_3, l_4, EE_x, EE_y] \\
 &\text{minimize}_{\mathbf{x}} && [f_\eta(\mathbf{x}), f_{\tau_1}(\mathbf{x}), f_{\tau_2}(\mathbf{x})] \\
 &\text{subject to} && lb \leq \mathbf{x} \leq ub \\
 & && g_1(\mathbf{x}) : f_\eta(\mathbf{x}) > 0 \\
 & && g_2(\mathbf{x}) : f_{\tau_1}(\mathbf{x}) > 0 \\
 & && g_3(\mathbf{x}) : f_{\tau_2}(\mathbf{x}) > 0 \\
 & && g_4(\mathbf{x}) : l_3 + l_2 < l_1 + l_4 \\
 & && g_5(\mathbf{x}) : f_\eta(\mathbf{x}) > \text{mean}[\eta(\mathbf{x})] - z \cdot \text{std}[\eta(\mathbf{x})] \\
 & && g_6(\mathbf{x}) : f_\eta(\mathbf{x}) < \text{mean}[\eta(\mathbf{x})] + z \cdot \text{std}[\eta(\mathbf{x})] \\
 & && g_7(\mathbf{x}) : f_{\tau_1}(\mathbf{x}) > \text{mean}[\tau_1(\mathbf{x})] - z \cdot \text{std}[\tau_1(\mathbf{x})] \\
 & && g_8(\mathbf{x}) : f_{\tau_1}(\mathbf{x}) < \text{mean}[\tau_1(\mathbf{x})] + z \cdot \text{std}[\tau_1(\mathbf{x})] \\
 & && g_9(\mathbf{x}) : f_{\tau_2}(\mathbf{x}) > \text{mean}[\tau_2(\mathbf{x})] - z \cdot \text{std}[\tau_2(\mathbf{x})] \\
 & && g_{10}(\mathbf{x}) : f_{\tau_2}(\mathbf{x}) < \text{mean}[\tau_2(\mathbf{x})] + z \cdot \text{std}[\tau_2(\mathbf{x})]
 \end{aligned} \tag{9}$$

To design a feasible quasi-serial manipulator, several constraints were derived as follows:

- (1) The kinematic performance (η) of the manipulator, which includes the target task, must be positive to ensure operational feasibility.
- (2) The required joint torques (τ_1 and τ_2) must also be positive, as negative torque values are physically infeasible in this design context.
- (3) The quasi-serial mechanism should satisfy the crank-rocker geometry constraint (in Eq. (1)), ensuring the linkage lengths allow proper mechanical motion.
- (4) To address singularity issues in the quasi-serial mechanism, designs that deviate significantly from the mean performance values are excluded. A z-value of 1.95, assuming a normal distribution, is used to filter out outliers for kinematic performance (η) and joint torques (τ_1 and τ_2). These constraints ensure that the performance metrics remain within a range defined by their mean and standard deviation.

The optimization problem for the quasi-serial manipulator can thus be summarized, as defined in Eq. (9), to minimize the kinematic performance and joint torque objectives while satisfying the above constraints. Each design variable $\mathbf{x}$ is further restricted to the lower and upper bounds specified in Table 1, ensuring all solutions remain within the feasible design space.

4. Optimization results

4.1. Dynamic analysis results

Dynamic simulations were conducted using RecurDyn 2023 (FunctionBay FunctionBay, 2023), analyzing 11,080 mechanisms over approximately 20 days. Each dynamic simulation required about 2.5 min, and including the time for parametric model export, approximately 3 min were needed per design candidate. The dynamic simulation automation process is summarized in Fig. 4: (1) importing the manipulator parts generated in Stage 1, (2) automatically assembling parts based on quasi-serial mechanism parameters by calculating joint and fixed parts, (3) assigning ABS-M30 material properties for 3D printing, (4) inputting the motion range of the frame and crank and specifying the payload, and (5) performing RBD analysis to obtain the required joint torques ($\tau_{1,2}$).

To simplify the analysis and avoid the computational burden of full-scale evaluations, the dynamic simulations were streamlined to focus on obtaining essential performance metrics relevant to the proposed problem. Although the resulting dataset of 11,080 labeled samples is extensive, such large-scale data generation is often impractical for real-world design problems due to excessive computational cost. Therefore, this dataset served as a reference for determining appropriate dataset sizes and guiding the DoE strategy. In the training process, only a subset of the dataset selected through adaptive sampling was utilized, ensuring computational efficiency without compromising model accuracy.

4.2. Surrogate model training results

The initial set of 200 design candidates was selected, and the AutoML process determined the optimal hyperparameters for the application. The identified hyperparameters were: hidden layer configuration (256, 128, 64, 32) and a learning rate of 0.01. Based on these tuned hyperparameters, the surrogate model was further refined through adaptive sampling to enhance its accuracy. As more samples were added, the optimal hyperparameters could change, requiring model re-evaluation. New sampling strategies and hyperparameter tuning could be applied if the desired performance was not achieved after adaptive sampling (Fig. 5). Including the adaptive sampling iterations and ensemble model construction, the total training time was approximately 12 min.

In each iteration of the adaptive sampling process, 50 new samples were added to the training dataset. These samples were prioritized based on regions of high uncertainty, as measured by the variance in predictions among the ensemble models. Specifically, the uncertainty for each design candidate was calculated as the standard deviation of predictions from the five surrogate models in the ensemble. The selection process focused on test set inputs ($\mathbf{x}$) with the highest variance in ensemble predictions, ensuring that the most uncertain regions were targeted for improvement. This iterative process continued until the target R^2 value of 0.95 or higher was achieved for all performance metrics. As a result, the adaptive sampling process converged with 2150 design candidates, producing a model with test R^2 values of [0.9984, 0.9828, 0.9833] and training R^2 values of [0.9987, 0.9531, 0.9558]. These results demonstrate the effectiveness of the adaptive sampling strategy in efficiently improving model accuracy.

Following the completion of the adaptive sampling process, an ensemble model was constructed using the refined hyperparameters and the final set of 2150 design candidates. The ensemble model was built by training five independent surrogate models, with the predictions aggregated based on R^2 -weighted averaging. The performance metrics for each of the five models and the final ensemble model are presented in Table 2. Additionally, the table includes the mean-variance of predictions across design variables for each model, highlighting the robustness of the ensemble model compared to individual surrogate models. The standard deviation of the outputs from the five models used to construct the ensemble model was 0.0021 (η), 7.7351 (τ_1), and

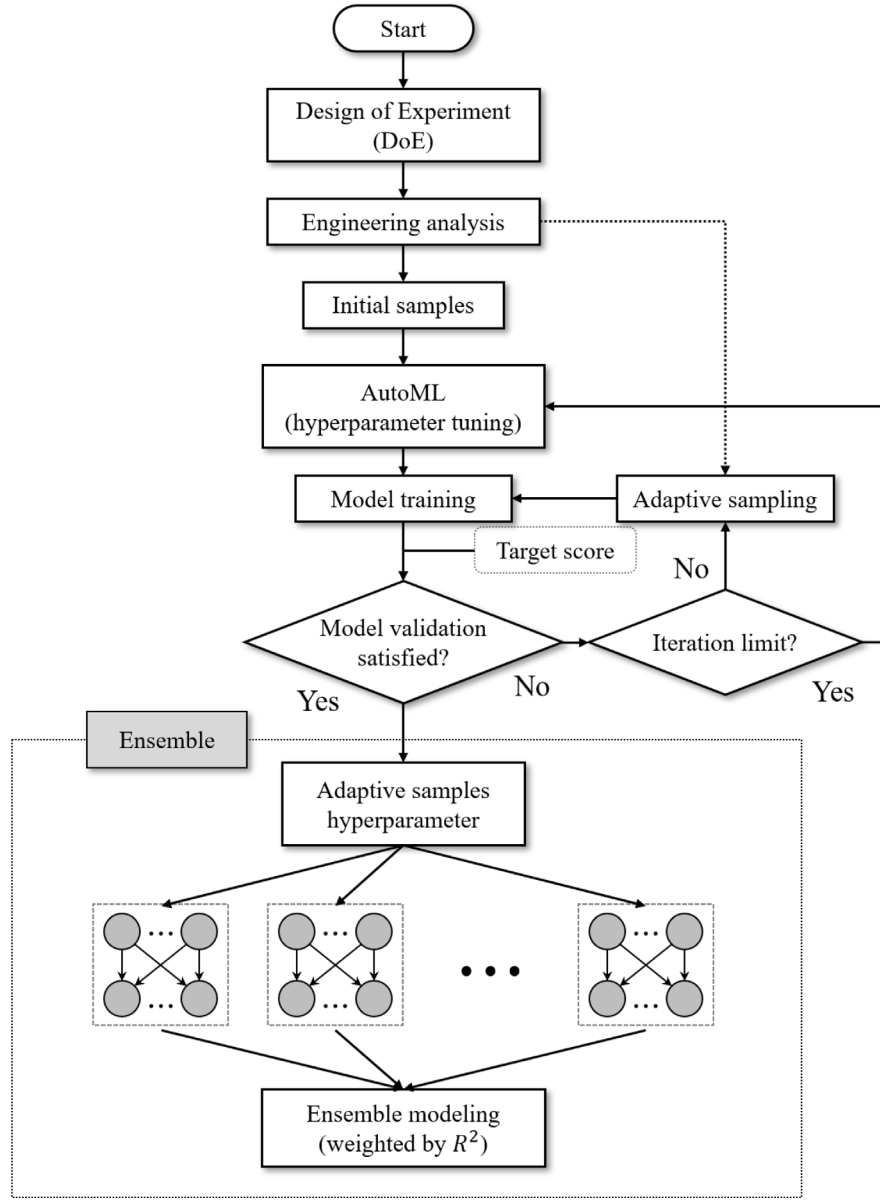


Fig. 5. Workflow of the proposed surrogate modeling framework for engineering performance prediction. The process begins with an initial dataset for training and validation, followed by hyperparameter tuning using AutoML. Adaptive sampling iteratively refines the surrogate model by adding informative samples, while ensemble modeling integrates multiple models weighted by their R^2 scores to enhance robustness and accuracy.

2.7161 (τ_2) for the respective target variables. Considering that the range of the dataset labels spans 0–0.14 (η), 100–600 (τ_1), and 40–160 (τ_2), these deviations are acceptably low, indicating that the models exhibit minimal variability in their predictions.

Fig. 6 compares the training and testing datasets' predicted and ground truth values. The R^2 values for the test dataset were occasionally higher than those for the training dataset due to the improved design space coverage achieved through adaptive sampling. This result confirms that adaptive sampling effectively enhances the model without overfitting the training data. The predictive performance of each model was evaluated using root mean squared error (RMSE) and mean absolute error (MAE), which are defined as follows:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}, \quad (10)$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|,$$

where y_i and $\hat{y}_i$ denote the ground truth and predicted values, respectively, and N is the total number of samples. By comparing the RMSE and MAE values (Eq. (10)) of the five MLP models and the ensemble model presented in Table 2, it is observed that the ensemble model consistently achieves lower error values. The arithmetic mean RMSEs across the objectives for the individual models were calculated as follows: train 0.0144 (η), test 0.0141 (η), train 0.0462 (τ_1), test 0.0275 (τ_1), train 0.0552 (τ_2), and test 0.0333 (τ_2). Similarly, the arithmetic mean MAEs were: train 0.0115 (η), test 0.0112 (η), train 0.0314 (τ_1), test 0.0177 (τ_1), train 0.0370 (τ_2), and test 0.0230 (τ_2). All of these values are higher than the corresponding RMSE and MAE values of the ensemble model shown at the bottom of Table 2, confirming the effectiveness of the R^2 -weighted averaging strategy.

Ensemble vs single MLP

To further evaluate the effectiveness of adaptive sampling, a comparison was conducted against a single MLP model trained using the entire dataset of 11,080 samples, split into an 8:2 ratio for training

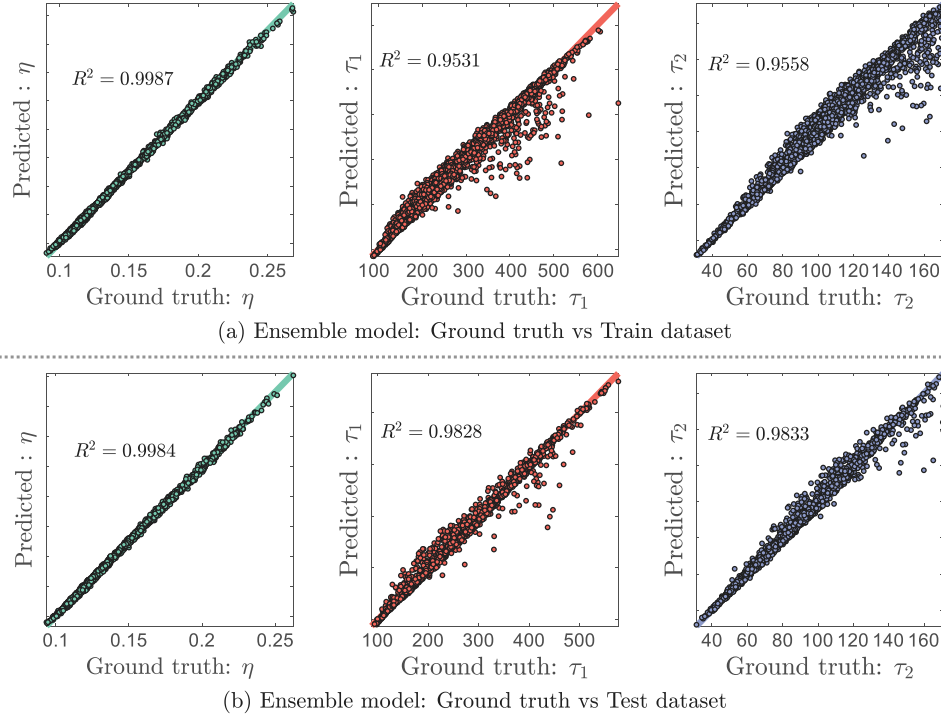


Fig. 6. Performance evaluation of the ensemble model for kinematic and dynamic metrics. A total of 2150 samples were used, split into training and test sets at an 8:2 ratio. (a) Comparison of predicted and ground truth values in the training set for η , τ_1 , and τ_2 . (b) Same comparison in the test set.

Table 2

Performance comparison of individual MLP models and the ensemble model.

Model		η		τ_1		τ_2	
		Train	Test	Train	Test	Train	Test
MLP ₁	R^2	0.9969	0.9957	0.9491	0.9777	0.9418	0.9663
	RMSE	0.0117	0.0116	0.0466	0.0258	0.0555	0.0363
	MAE	0.0092	0.0092	0.0326	0.0167	0.0421	0.0274
MLP ₂	R^2	0.9970	0.9960	0.9215	0.9692	0.9286	0.9710
	RMSE	0.0114	0.0113	0.0579	0.0304	0.0615	0.0337
	MAE	0.0089	0.0089	0.0353	0.0186	0.0379	0.0214
MLP ₃	R^2	0.9899	0.9893	0.9481	0.9689	0.9459	0.9734
	RMSE	0.0211	0.0184	0.0471	0.0305	0.0535	0.0323
	MAE	0.0185	0.0159	0.0328	0.0223	0.0367	0.0214
MLP ₄	R^2	0.9949	0.9921	0.9650	0.9847	0.9607	0.9766
	RMSE	0.0151	0.0159	0.0387	0.0214	0.0456	0.0303
	MAE	0.0111	0.0120	0.0229	0.0131	0.0314	0.0219
MLP ₅	R^2	0.9964	0.9945	0.9258	0.9715	0.9317	0.9704
	RMSE	0.0126	0.0133	0.0563	0.0292	0.0601	0.0340
	MAE	0.0099	0.0102	0.0335	0.0177	0.0367	0.0227
Ensemble	R^2	0.9987	0.9984	0.9531	0.9828	0.9558	0.9833
	RMSE	0.0075	0.0072	0.0447	0.0227	0.0483	0.0256
	MAE	0.0058	0.0058	0.0263	0.0131	0.0296	0.0144

and testing. The results, shown in Fig. 7, indicate that although no significant difference in R^2 values is observed between the models in Figs. 6 and 7, the adaptive sampling approach achieved comparable predictive performance using approximately five times fewer samples (2150 vs 11,080). This result highlights the robustness and efficiency of the ensemble model constructed through adaptive sampling, confirming that high prediction accuracy can be maintained even with a significantly reduced dataset size. Assuming that the parametric automation process, including model export and RBD analysis, requires approximately three minutes per design candidate, the adaptive sampling approach results in an estimated reduction of about 447 h of computational time. Despite the relatively fast nature of RBD analysis, this reduction is substantial. Furthermore, it is expected that the

time savings achieved through adaptive sampling would be even more pronounced for more computationally intensive analyses, such as finite element analysis (FEA) or evaluations involving thermal or vibrational mechanical properties. Overall, this resulted in a significant reduction in computational cost for the proposed application, decreasing the required dataset size by approximately five times while maintaining excellent predictive performance.

4.3. Evaluation of pareto solutions in multi-objective optimization

Based on the trained ensemble model, NSGA-II was conducted using the optimization formulation defined (in Eq. (9)), and Fig. 8 displays the resulting Pareto front. Since the relative lengths for each frame were used when defining design variables in Section 3.1, the effects of the scale factor, which approximates the size of the manipulator, were analyzed. In general, the optimized solutions indicated that as the scale factor increased, η decreased while $\tau_{1,2}$ increased. This confirms that the size of the manipulator significantly influences both the workspace and joint torque, even in the optimal solutions. For a more intuitive comparison, the manipulator and workspace for each section are visualized in Fig. 8. The final design to be realized using the 3D printer is represented as a diamond symbol in Fig. 8, selected based on the maximum manufacturable size. The 3D printer used in this study has an output size of $W \times L \times H$: 600 mm $\times$ 350 mm $\times$ 350 mm.

In addition to the Pareto front obtained via NSGA-II, several optimal solutions derived using the SQP method are also highlighted in Fig. 8. These solutions were obtained by applying the SQP algorithm to the trained ensemble model, where the multi-objective problem was converted into a series of single-objective formulations through weighted sums of the three objectives. Eight SQP optimal solutions were considered by assigning weights of 1.0 or 2.0 to each objective. The results show that most of the SQP-derived solutions lie in the region of the Pareto front corresponding to larger scale factors, indicating that the trade-offs implicitly embedded in the weighted sum formulation tend to favor larger manipulators.

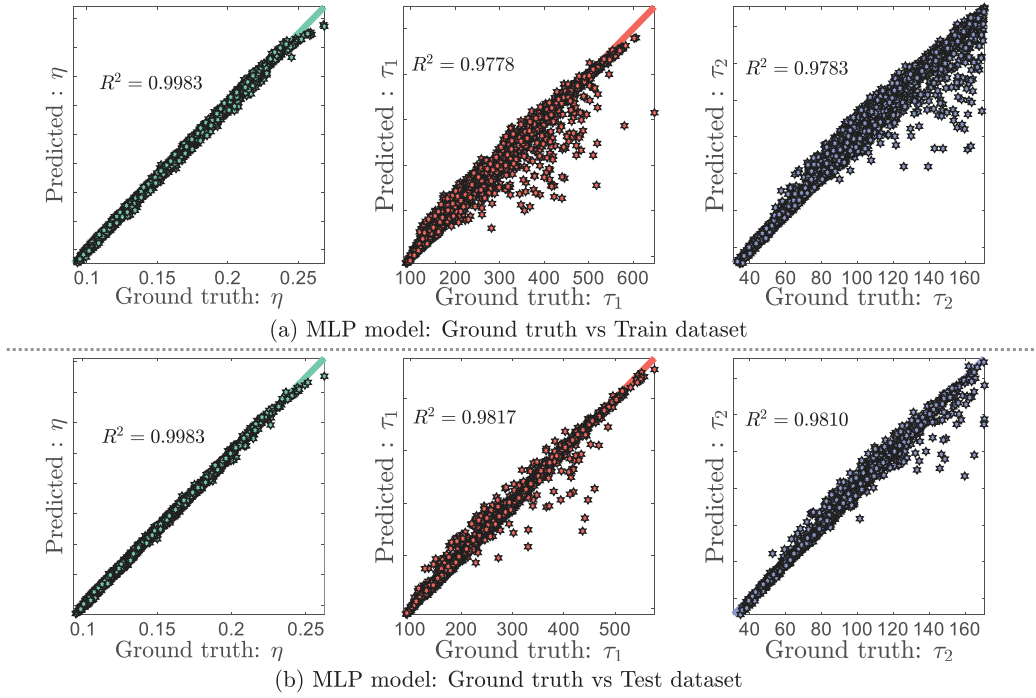


Fig. 7. Performance evaluation of the single MLP model for kinematic and dynamic metrics. A total of 11,080 samples were used, split into training and test sets at an 8:2 ratio. (a) Comparison of predicted and ground truth values in the training set for η , τ_1 , and τ_2 . (b) Same comparison in the test set.

Although the weighted-sum approach using SQP can yield solutions similar to those on the Pareto front, it may be less effective for early-stage design exploration, where visualizing the diversity of feasible manipulator sizes and configurations under manufacturing constraints is critical. This comparison illustrates the advantage of NSGA-II in capturing a broader set of trade-off solutions, especially when dealing with multi-faceted design requirements.

A comparison between the predicted and ground truth values for the Pareto solutions was conducted to evaluate the reliability of the trained surrogate model. Fig. 9 shows the R^2 performance of the surrogate model predictions against the ground truth values obtained from dynamic simulations. The results show high prediction accuracy for the payload torques (τ_1 , τ_2), while the accuracy for the utile ratio (η) was relatively lower. Nonetheless, the predicted η values generally captured the overall trend of the ground truth values, indicating that the model maintained a reasonable level of reliability.

The high prediction accuracy for τ_1 and τ_2 can be attributed to their labeling method. These torque values are determined based on the maximum payload torque observed within the reachable workspace of the end-effector via dynamic analysis. As such, they tend to be less sensitive to slight variations in the design variables, making them easier for the surrogate model to learn and predict. In contrast, the relatively lower prediction performance for η can be explained by its higher sensitivity to the design variables. Since η represents the ratio of the task region to the reachable workspace, objective space changes significantly with even slight variations in design variables. This heightened sensitivity stems from the nonlinear nature of how the workspace deforms in response to geometric changes, especially when determining whether the task region is fully enclosed within the workspace. Moreover, in quasi-serial mechanisms, it is inherently difficult to achieve a workspace that perfectly fits the task region, as residual unused areas are unavoidable. Since η is defined as the ratio of the task region to the overall workspace, minimizing η effectively encourages the design of manipulators with wider workspaces while maintaining the ability to perform a given task. This formulation is beneficial because minimizing η encourages the design of manipulators that cover the given task region with a larger workspace, providing

more flexibility in similar tasks without compromising task region coverage. However, this also makes η more sensitive to geometric changes, especially near the optimum, where slight design variations can significantly alter the effective workspace. As a result, the prediction of η becomes more nonlinear and less stable than other performance metrics, contributing to relatively lower accuracy in estimating η . The next step is to examine how each design variable influences the performance metrics. This analysis, detailed in Section 5, emphasizes variables with high sensitivity that could impose limitations during the subsequent detailed design stage, and identifies critical thresholds that should be considered to maintain desirable performance.

5. Design rule extraction

Several additional analyses were performed to extract design rules for quasi-serial mechanisms using the trained ensemble model, which enables real-time prediction of arbitrary designs. First, in Section 5.1, a sensitivity analysis employing the Sobol approach was conducted across the entire design space to evaluate the importance of design parameters with respect to each objective (Sobol, 2001; Herman and Usher, 2017). Next, in Section 5.2, random forest was applied to the design space near the Pareto solutions to extract multivariate design rules. Lastly, in Section 5.3, Spearman and Pearson correlation analyses were conducted for further investigation near Pareto solutions, with details of the applied data mining techniques available in previous studies (Yang and Yee, 2022).

5.1. Sobol sensitivity analysis

A total of 14,336 design points were sampled on the trained ensemble model using a Sobol sampler within the predefined design space; for this purpose, the Python library SALIB was utilized. The importance of design variables ($\mathbf{x}$) on the objective values (η , $\tau_{1,2}$) was investigated through Sobol sensitivity analysis, and the results are presented in Fig. 10. For η , EE_x , l_3 , and l_1 were identified as the most significant contributors in that order (Fig. 10(a)). The EE_x variable was rated as the most important factor because it is the design variable closest to the

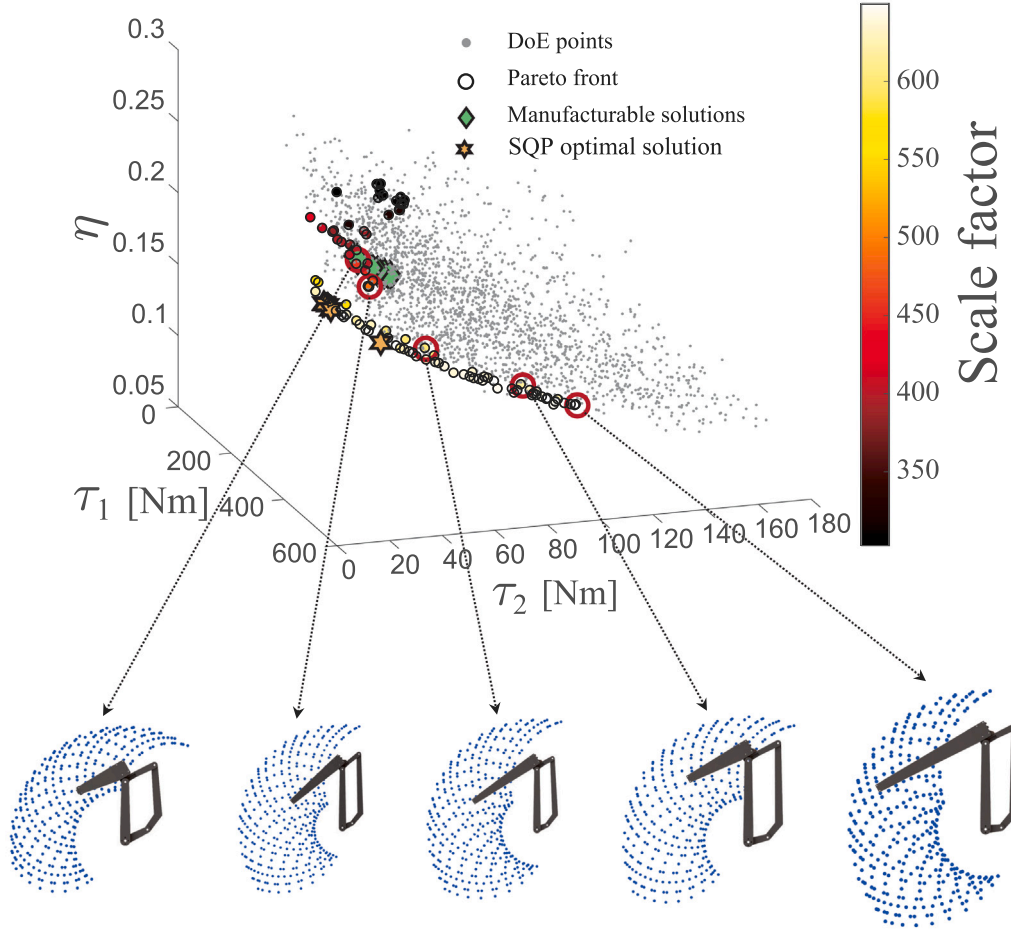


Fig. 8. Pareto solutions obtained by NSGA-II and configuration of 5 selected designs.

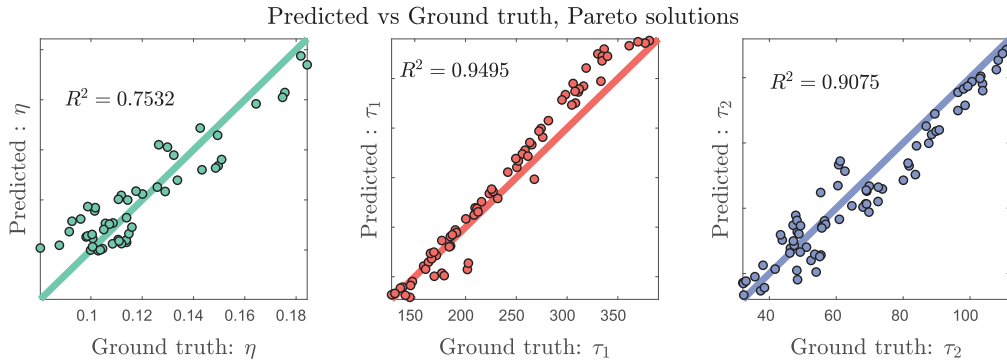


Fig. 9. Comparison between predicted and ground truth performance values for Pareto solutions.

end-effector. l_3 was ranked second, as it influences the range of rotation for the upper arm. Lastly, l_1 was ranked third due to its association with the rotational position of the upper arm. Since the workspace is defined as the range of the end-effector reachable through rotation, it can be concluded that the objective function η is highly sensitive to changes in the rotational range. In contrast, it is relatively insensitive to l_2 , l_4 , and EE_y , which primarily affect position rather than rotation. To verify these observations quantitatively, the position of the end-effector was mathematically derived in terms of the design variables, as shown below.

First, the position of the frame (l_1) is determined by θ_1 and the position of EE_y is determined by the kinematic condition of the quasi-serial mechanism (in Fig. 11). Then, the distance a between B and C can be obtained using the Cosine law as Eq. (11).

$$a = \sqrt{l_1^2 + l_2^2 - 2l_1l_2 \cos(\theta_1 - \theta_2)} \quad (11)$$

The angle $\angle ABD$ between l_1 and l_4 can be computed by the sum of ζ and ξ , where ζ and ξ are defined as follows:

$$\zeta = \cos^{-1}\left(\frac{l_1^2 + a^2 - l_2^2}{2l_1a}\right), \quad (12)$$

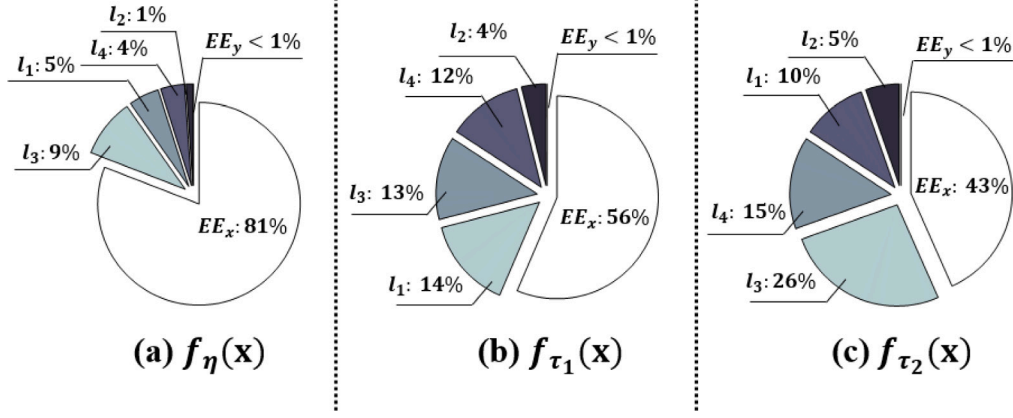


Fig. 10. Sobol results for each objective function.

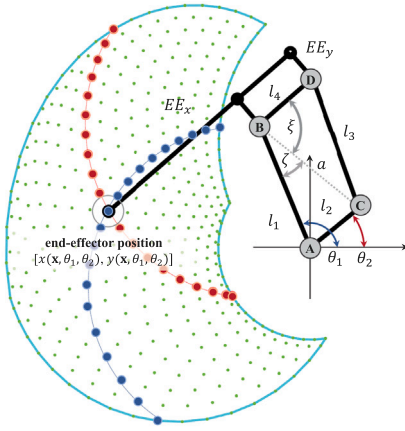


Fig. 11. Kinematic configuration of the quasi-serial mechanism.

$$\xi = \cos^{-1}\left(\frac{l_4^2 + a^2 - l_3^2}{2l_4a}\right). \quad (13)$$

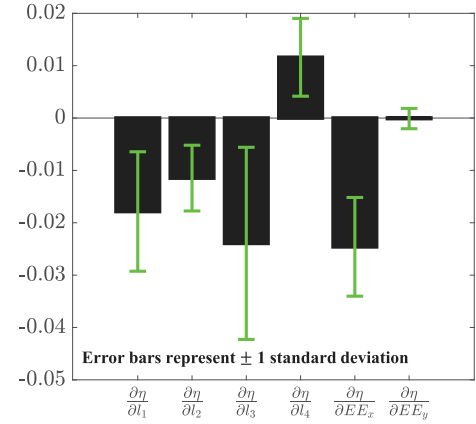
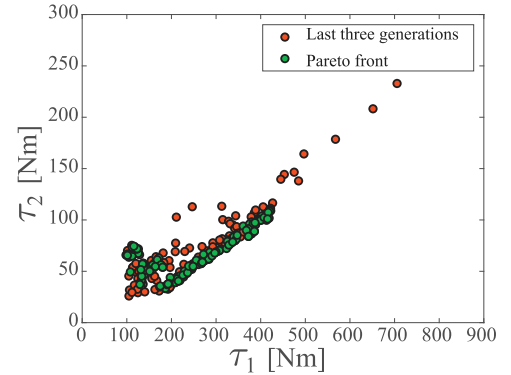
Then, the above relation allows one to express the end-effector position $[x, y]$:

$$x(\mathbf{x}, \theta_1, \theta_2) = l_1 \cos(\theta_1) + EE_y \cos(\theta_1 + \zeta + \xi - \frac{\pi}{2}) + EE_x \cos(\theta_1 + \zeta + \xi), \quad (14)$$

$$y(\mathbf{x}, \theta_1, \theta_2) = l_1 \sin(\theta_1) + EE_y \sin(\theta_1 + \zeta + \xi - \frac{\pi}{2}) + EE_x \sin(\theta_1 + \zeta + \xi). \quad (15)$$

The derived coordinates of the end-effector were utilized to analyze their variation with respect to the design variables $\mathbf{x}$ across the 11,080 sample designs generated in Section 3.1. To achieve this, derivatives of η with respect to the six design variables were computed for all 11,080 designs, and the statistical results are summarized in Fig. 12. Notably, the derivative values were found to be largest for EE_x , followed by l_3 and l_1 , aligning with the results obtained from the Sobol sensitivity analysis.

Back to the Sobol analysis, the results of τ_1 and τ_2 are then discussed. Note that since these torques are affected by various properties (such as mass, inertia) of each linkage, it is difficult to construct a mathematical model as in η . In Fig. 10(b), it can be found that τ_1 has similar important design variables to η : EE_x , l_1 , and l_3 reappear but the order between l_1 , and l_3 changes. Again, EE_x , the variable closest to the workspace, was evaluated as the most influential design variable. Then, l_1 appeared, which is closely related to τ_1 in that it is the rod where the torque is applied directly. The l_3, l_4 were ranked third and fourth without a significant difference: they constrain the coupling to 1-DOF while determining the upper arm's range of motion. Thus, it

Fig. 12. First derivative of kinematic performance (η) with respect to design variables.Fig. 13. Scatter plot of τ_1 and τ_2 : Pareto solutions (green) and populations from the last three generations of NSGA-II (orange).

was evaluated to have a similar effect on $\tau_{1,2}$, which affects the upper arm. For τ_2 (Fig. 10(c)), EE_x has the most influence again. However, l_4 appears as the next important variable with l_3 . As in the τ_1 , they are analyzed to have similar importance due to their role in structural robustness. These results are consistent with the properties of the quasi-serial manipulator studied previously, where it has the advantage of structural efficiency and higher rigidity with a higher payload than a serial manipulator with the same motor torque (Klimchik et al., 2016b; Klimchik and Pashkevich, 2017; Klimchik et al., 2016a). In summary, given that the design variables extracted from the Sobol

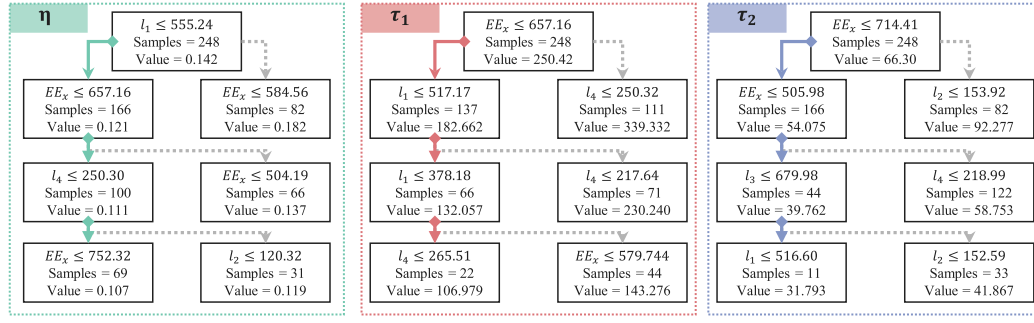


Fig. 14. Random forest results for each objective function.

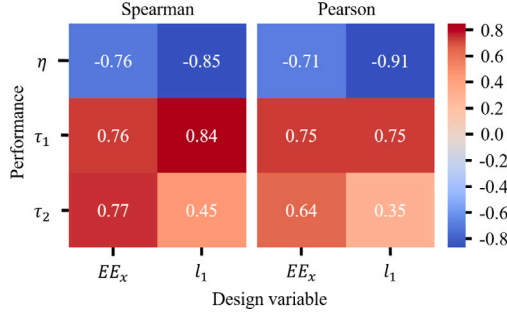


Fig. 15. Spearman and Pearson correlations between design variables and objective functions: only significant relations are presented.

analysis are analyzed to have a reasonable underlying physics for their impact on the objective functions, it can be concluded that the trained ensemble model has good enough global accuracy to be used for design optimization.

5.2. Random forest

Random forest is an ensemble learning method that constructs multiple decision trees and aggregates their outputs to improve predictive performance and robustness (Breiman, 2001). Due to its ability to handle non-linear relationships and assess variable importance, it has been widely adopted for feature selection and model interpretation in engineering applications. Employs random forest analysis to investigate the trade-off relationships between the three objective functions, complementing the Sobol analysis discussed in the previous section. There are 400 design samples from the trained ensemble surrogate model, including 100 Pareto solutions and 300 additional samples from the populations of the last three generations in NSGA-II (Fig. 13). To more comprehensively explore the design space near the Pareto front, 62% of the samples, corresponding to 248 designs, were randomly selected for the random forest analysis (Yang and Yee, 2022).

The results of the random forest analysis are shown in Fig. 14. Bold arrows indicate the direction of optimization; for instance, the arrows in the random forest for η point toward maximizing η . For $\tau_{1,2}$, EE_x consistently appears as the primary influencing variable, consistent with the Sobol analysis results. However, the random forest provides further information by indicating the preferred adjustment direction for each objective function. Notably, the analysis reveals a common design rule that decreasing l_1 and EE_x improves all three objectives. The first branch of the random forest for η is split at the threshold $l_1 \leq 555.24$, a finding further supported by Pearson correlation coefficients showing a strong negative correlation between η and l_1 (−0.85 and −0.91) (Fig. 15). While the Sobol sensitivity analysis identified EE_x as a key influencing factor, it did not indicate the direction of adjustment. In contrast, the random forest analysis provides more actionable insights

by revealing the trade-off relationship between η and τ_1 : increasing l_1 benefits τ_1 but deteriorates η . Thus, random forest analysis identifies critical variables and offers directional guidance, aiding more informed design adjustments in MOO.

5.3. Correlation coefficients

This subsection focuses on the analysis of correlation coefficients to intuitively substantiate the findings from the preceding random forest analysis. This examination provides a comprehensive understanding of the relationships between the input and output variables. To accomplish this, both Spearman and Pearson correlation coefficients were computed for 400 designs, as described in Section 5.2. The results, shown in Fig. 15, indicate that only l_1 and EE_x are statistically significant with respect to the outputs.

Both the Spearman and Pearson correlation analysis reveal that EE_x has negative correlations with η while showing positive correlations with τ_1 and τ_2 ; these results are physically reasonable that the required torque increases as the workspace expands and the length of the upper arm increases. Also, they are in perfect agreement with the previous Sobol and random forest analyses. Sobol denoted that EE_x is always the most dominant variable for all objectives and random forest denoted that EE_x should be decreased to maximize η and minimize τ_1 and τ_2 . However, given that the Pearson coefficient indicates that the two variables are linearly correlated, the results of Pearson analysis further indicate that the variable EE_x has a linear relationship with the three objectives. Moreover, Pearson coefficient between l_1 and η is found to be linearly correlated with negative direction. It is also consistent with the random forest of η : the second and third branches of it indicate that l_1 should decrease to maximize η . The reason for the negative linear relationships between two design variables (l_1 and EE_x) and the objective η is that the workspace becomes wider as the length of the upper and lower arms increases: this can be verified more intuitively in Fig. 16. The length of l_1 is the scale factor, and the lengths of the other variables are determined as a ratio to l_1 . The scale factor does not indicate the characteristics of each mechanism, but rather the overall size of the manipulator. As shown by the gradation in Fig. 16, there is a significant variation between manipulators with similar scale factors. Thus, it can be concluded that performance depends more on the structure of the mechanism rather than the size of the manipulator required to perform the target task. This suggests that design optimization of quasi-serial mechanisms is essential.

5.4. Summary of design guidelines

Notably, EE_x and l_1 , which correspond to the dimensions of the lower and upper arms, are typical design parameters for serial-type manipulators, and their importance is intuitively expected. Interestingly, these variables were also found to be the most influential in the quasi-serial manipulator, highlighting that the lower and upper arms remain the most critical components even across structurally similar but distinct manipulator architectures. Another noteworthy finding concerns

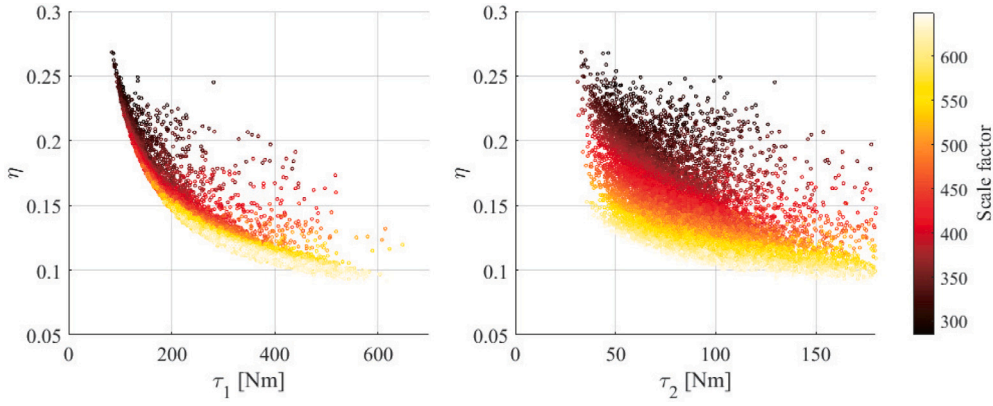


Fig. 16. Engineering performances of 11,080 manipulator designs colored by scale factor.

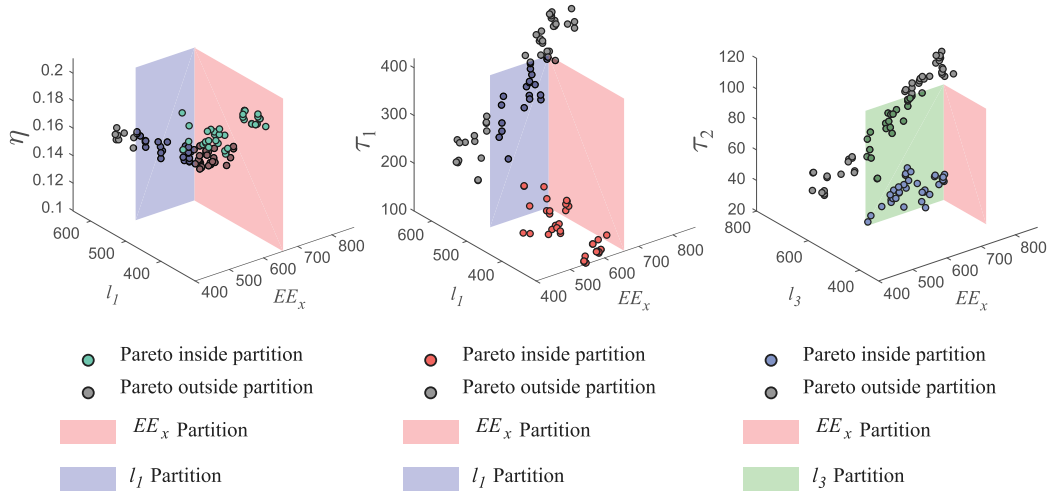


Fig. 17. Visualization of Pareto solutions within the design space partitioned by random forest. The partitions are defined based on critical design variables (EE_x , l_1 , and l_3) to limit the design space and identify regions requiring additional DoE or evaluation of target performance metrics.

the relative importance of l_3 (coupler) and l_4 (rocker). Traditionally, adjusting l_4 has been considered a more straightforward way to meet workspace or payload torque requirements due to its direct influence on the end-effector trajectory. However, the current results suggest that modifying l_4 alone can lead to infeasible designs that violate practical constraints, particularly in extreme configurations where Eq. (1) is marginally satisfied. This highlights that the sensitivity of l_3 originates from its geometric role in defining the coupler curve. Due to the high sensitivity and nonlinearity of the coupler motion, modifying l_3 has traditionally been significantly challenging. As a result, l_4 is more commonly adjusted in practice, as it is more intuitive and stable. Nevertheless, the present analysis reveals the overlooked design potential of l_3 , indicating its critical impact on workspace and payload torque optimization.

These findings from the sensitivity analysis conducted in the previous sections, where Sobol analysis quantified the relative importance of each variable, and random forest analysis provided comprehensive insights into the directional influence of these variables on the objectives. The partitioned design space based on these key variables is visualized in Fig. 17, where the boundaries determined by the random forest analysis effectively segment the space to highlight regions of interest.

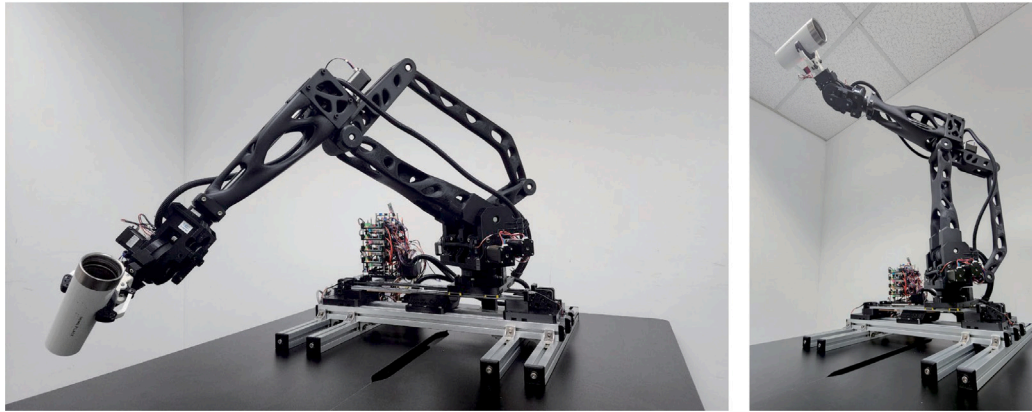
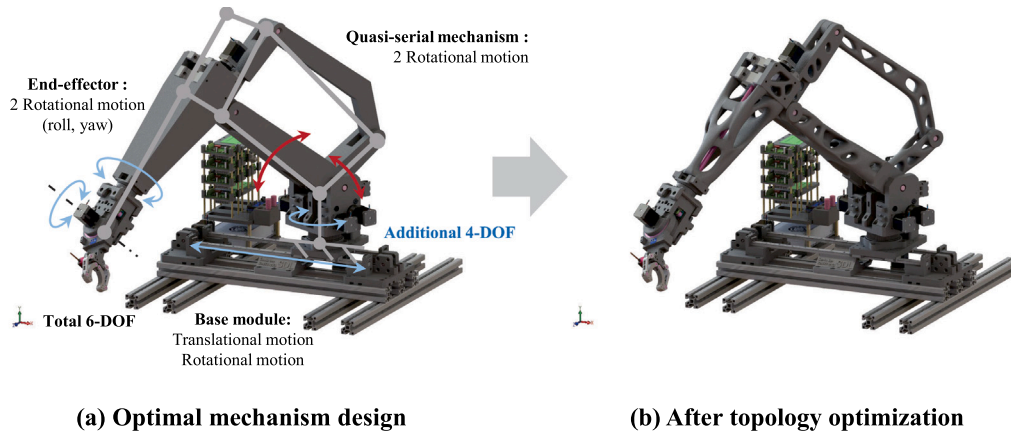
The extracted design rules and guidelines help identify critical variables for performance prediction and support the formulation of adaptive sampling strategies. These rules can selectively augment training data in underrepresented or low-performing regions of the design space and refine the sampling focus on sensitive variable ranges. Additionally, they can aid in evaluating optimal solutions obtained through

algorithms such as NSGA-II or SQP by indicating how many of the derived designs fall within feasible and meaningful parameter regions. This provides practical guidance for interpreting optimization outcomes and navigating future design iterations more efficiently.

6. Realization of the optimal quasi-serial manipulator

In this section, the embodiment design of the quasi-serial manipulator is described, including its construction and actuation. A total of 100 optional concept quasi-serial manipulator candidates were obtained using the proposed framework. Candidates with appropriate sizes, suitable for manufacturing via 3D printing, were selected (Fig. 8). The design focused on a 2-DOF mechanism, which has a major impact on the quasi-serial manipulator's ability to perform the target task. However, an actual manipulator typically requires 6-DOF. Therefore, the additional 4-DOF was designed during the embodiment design step. Two DOFs were added to the base to allow the manipulator to rotate and translate, and roll and yaw DOFs were added to the end-effector to enable object grasping at various angles and positions (Fig. 18(a)). To implement a feasible design, the configuration of motors, gears, bearings, controllers, and other components was also considered. Additionally, topology optimization was performed to further improve the dynamic performance of the recommended design (Fig. 18(b)). Finally, the components were assembled to realize the actual quasi-serial manipulator (Fig. 18(c)).

The required actuator specification before and after topology optimization for payload (5 kg) is the same as in Table 3. The motor used to



(c) Realization of the quasi-serial manipulator

Fig. 18. Embodiment design and topology optimization.

achieve the required actuator specification is a NEMA17 stepper motor designed for a maximum output of 150 Nm and 100 Nm using a 50:1 gear and timing belt. A typical motor has a characteristic where output angular velocity and torque are inversely proportional. Therefore, to output the same torque as shown in Table 3, the required torque was satisfied by adjusting the speeds of the stepper motor. Experiments were also conducted to perform the tasks shown in Fig. 2. The payload used in the dynamic simulation was 5 kg, but the weight of the gripper added during the embodiment design process was about 2 kg. Therefore, the 3 kg object was gripped and the task was performed (Fig. 19).

The results of the payload test confirmed that the quasi-serial manipulator operated successfully under the task conditions defined in Fig. 2. These experimental results demonstrated that the proposed quasi-serial manipulator, including the task-specific design problem definition and realization process, can be effectively implemented. Although the manipulator presented in this study was not benchmarked against widely commercialized models, a series of ablation studies was conducted using non-adaptive sampling, adaptive sampling, NSGA-II, and SQP methods to evaluate the potential of the proposed problem formulation. The successful application of the optimization results to a newly defined task validates the practicality of the surrogate model-based optimization framework. This suggests that the proposed framework can be extended and applied to various other mechanism design applications through active problem formulation and optimization.

Table 3

Comparison of joint payload torque requirements before and after topology optimization.

	τ_1 [Nm]	τ_2 [Nm]
Optimal mechanism design	132.6011	37.63381
After topology optimization	98.58294	31.4224
Reduced ratio	25.65%	16.50%

7. Conclusion

This study presented a data-driven design framework to optimize the kinematic and dynamic performance of quasi-serial manipulators simultaneously. The feasibility and effectiveness of the proposed methodology were demonstrated through application to a quasi-serial manipulator, validating its potential for use in practical mechanism design problems.

Adaptive sampling was employed to iteratively update training data based on model uncertainty to enhance the predictive capability of the surrogate models while minimizing computational cost. Additionally, ensemble modeling using R^2 -weighted averaging improved prediction robustness across different mechanism configurations. These techniques collectively reduced the computational burden while maintaining high accuracy. A MOO process was conducted using the trained surrogate models. The global optimizer NSGA-II obtained Pareto solutions, efficiently exploring diverse trade-offs in the design space. In contrast, the local optimizer SQP, while capable of approximating individual Pareto points using weighted-sum formulations, was limited by the need to predefine weights. These results suggest that NSGA-II is more

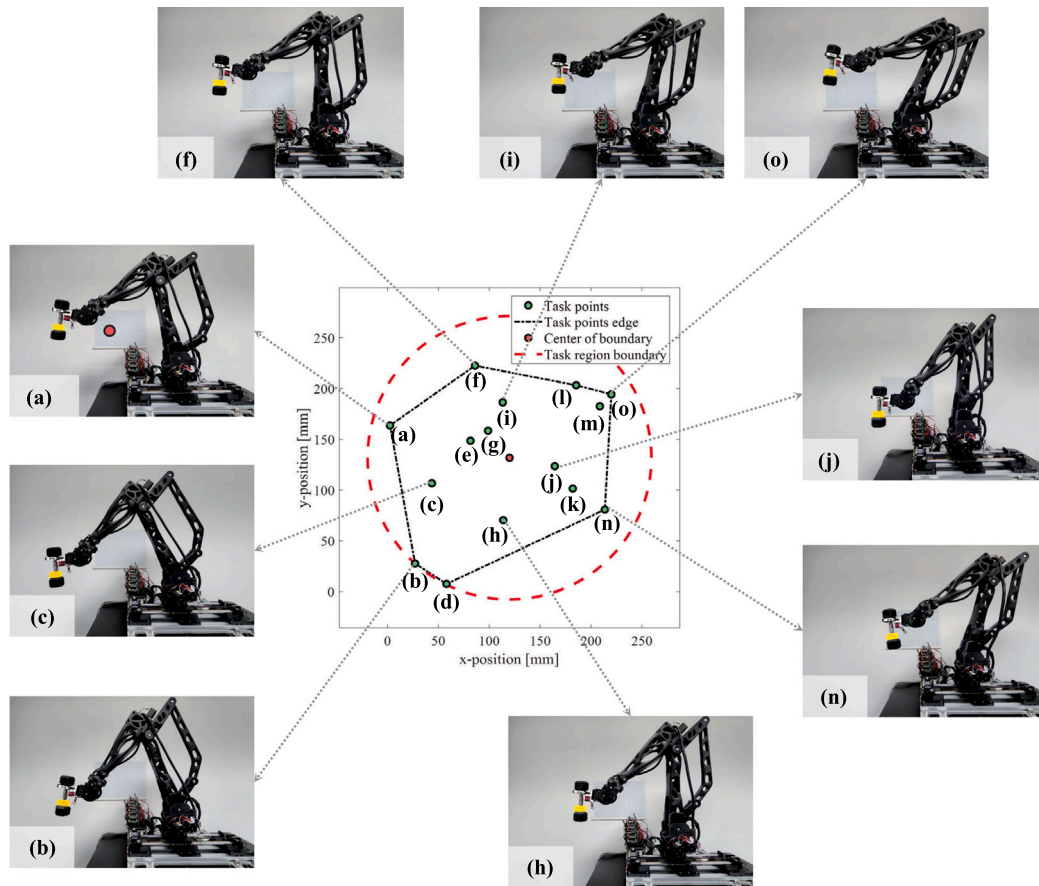


Fig. 19. Realized manipulator payload test actuation.

suitable for the proposed application. Sensitivity analyses using the Sobol method and random forest modeling further guided the design process by identifying key variables and their directional influence on performance metrics. These insights supported informed decisions in the conceptual and detailed design stages, including topology optimization for structural refinement and experimental validation through payload testing.

Before applying the proposed framework, parametric modeling and dynamic simulations were manually conducted for each candidate design. For example, conducting a sensitivity analysis on 14,336 designs would require approximately 717 h. By integrating adaptive sampling and ensemble modeling, the same task was completed in about 108 h, achieving an estimated 85% reduction in computational effort compared to traditional methods.

CRedit authorship contribution statement

Sumin Lee: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Sunwoong Yang:** Methodology, Formal analysis. **Namwoong Kang:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Replication of results

The code and data are available from the corresponding author on reasonable request.

Data availability

Data will be made available on request.

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